

Normal-flow virus filtration: detection and assessment of the endpoint in bioprocessing

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The breakthrough of a model virus, bacteriophage Φ X-174, through normal-flow virus filters was studied using both commercial process fluids and model feedstreams. The results indicate that (i) Φ X-174 is a reasonable model for a mammalian parvovirus [MMV (murine minute virus)] in virus filtration studies; (ii) Φ X-174 LRV [log (reduction value)] shows a better correlation with percentage flow decline compared with volume processed under a variety of conditions; (iii) although the extent of decline in virus LRV is dependent on the mechanism of filter fouling, the fouling mechanisms operative in a viral validation study are representative of those likely to be found under actual production conditions. The mechanism of LRV decline by many process streams was proposed to be due to selective plugging of small pores. A theoretical model as well as a predictive equation for LRV decline versus flow decay was derived; experimental results from filtration studies using pore-plugging feedstocks were consistent with the equation. As protein solutions may vary in their adsorptive versus plugging behaviour during filtration, an evaluation of the LRV-versus-flow-decay relationship on a biopharmaceutical-product-specific basis may be warranted.

Introduction

A majority of TPs (therapeutic proteins) and antibodies are currently expressed in mammalian cell lines at the manufacturing scale. Although alternative expression systems are being explored, it is likely that mammalian cells will continue to be used in the future, especially with proteins that require complex glycosylation patterns for their function. Instances of mammalian-cell-culture contamination with adventitious viruses have occurred in the past [1], and many mammalian cell lines produce endogenous retrovirus particles [2]. Regulatory expectations and good business practices dictate the adoption of measures to assure viral safety of protein pharmaceuticals [2a]. A multipronged

strategy is typically used to assure viral safety. The strategy includes spike/recovery experiments using model viruses to measure the viral clearance capacity of individual steps of the purification process. In early phase development, it is normally necessary to demonstrate the ability to remove or inactivate a model of endogenous retroviruses. For licensing, evaluation of two or three additional viruses is generally required [2a,2b].

A small quantity of a concentrated virus spike, typically less than 5 % of the volume of the process fluid, is spiked into a representative process feedstock. The concentration of the virus in the inlet and outlet streams is measured and the volume-corrected data are used to determine the viral-reduction ability of the process step and expressed as a log₁₀ reduction value {LRV; log ([virus in]/[virus out])}. Process, safety and current Good Manufacturing Practice (cGMP) considerations may dictate viral clearance reduction factors exceeding 10¹⁷–10¹⁸ for the entire purification process when cell substrates are used that produce high titres of type C endogenous retrovirus-like particles [e.g. CHO (Chinese-hamster ovary) cells and murine hybridomas].

Several methods are used for viral reduction including heat, chemical treatments [3], low pH [4], microwave heating [5], adsorption by chromatography [6] and mechanical/molecular sieving by filtration using normal- and tangential-flow methods [7–10]. The methods of choice depend on many factors, including product stability and physical properties of the target viruses. For example, enveloped viruses are generally easier to clear or inactivate than small non-enveloped viruses. In order to ensure the highest level of viral safety, it is important to understand the process

Key words: bioprocessing, log (reduction value), murine minute virus, normal flow virus filtration, pore-plugging model, virus breakthrough.

Abbreviations used: cGMP, current Good Manufacturing Practice; CPE, cytopathic effect; DMEM, Dulbecco's minimum essential medium; FBS, fetal bovine serum; LRV, log (reduction value); mAb, monoclonal antibody; MMV, murine minute virus; NFP, normal-flow parvovirus; p.f.u., plaque-forming units; TCID₅₀, 50 % tissue-culture infectious dose; TP, therapeutic protein; UF, ultrafilter.

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parameters that impact the efficacy of viral-inactivation or -removal steps.

The present investigation focuses on normal-flow viral filters and the phenomenon of virus breakthrough, which is observed under certain conditions of operation. It is generally regarded that the defined porous structure of virus filter membranes acts as a sieving barrier, where large virus particles (20–100 nm) are trapped while protein molecules (< 10 nm) pass through. Previous studies involving flux analysis have shown that, at some point of time during filtration, the near-complete exclusion breaks down and virus particles are detected in the permeate [8, 10–12]. Removal of viruses is the sole purpose of including a virus filter in a bioprocess, and unit operation robustness is a pivotal concern in bioprocessing. Fully validated unit operations are expected to perform within defined acceptance criteria from batch to batch and over the duration of an individual run. The virus breakthrough point is of great concern in bioprocessing for obvious safety reasons and is indicative of a decrease in the performance of the filter beyond that acceptable in commercial manufacturing. Thus unexpected breakthrough during the run is considered unacceptable by end users and regulatory agencies from a cGMP and validation standpoint.

In the present study, we examine the breakthrough of a model bacteriophage, Φ X-174, using Viresolve NFP (normal-flow parvovirus) filters and model process streams. Although the breakthrough of virus has been reported previously during validation trials [8, 10–12], it is unclear if this is also a concern for large-scale processing and what parameters contribute to it. The ultimate goal of this work is to develop strategies to avoid this phenomenon and to maximize the viral safety of biopharmaceutical products.

Theory

LRV versus flow-decline model

The model described below assumes that the mechanism of virus removal is based entirely on size exclusion as opposed to adsorptive effects. Filters possess a distribution of pore sizes and are effectively molecular sieves for particles of the size of viruses. Although the filter manufacturer's goal is to keep the pore-size distribution as defined and tight as possible, inevitably some variation occurs. The membrane has a large majority of pores smaller than virus particles and a small fraction of pores that are somewhat larger than viruses. It is hypothesized that the small pores are retentive to viruses and protein aggregates, and that the large pores make up a small proportion of the membrane surface and contribute insignificantly to the overall flow of liquid. However, as the filtration progresses, the small pores become progressively plugged by the protein aggregates and smaller particulates. As a result, the liquid flow through the small pores gradually decreases relative to flow through

the large pores, which is assumed to remain relatively constant. Consequently, a point is reached at which permeate flow through large pores is a relatively significant proportion of the overall liquid flux. Some of these pores are larger than viruses, and the alteration in pore flow distribution results in the experimentally observed breakthrough, lowering the LRV.

The starting point of the model development is the relationship between flow and sieving through the individual pores in the context of the overall flow and sieving through a membrane [13]. In this model, the pore-size distribution of the membrane is divided into two independent classifications, small pores and large pores, based on the size of a pore relative to a virus particle. Small pores are retentive to viruses and large pores allow virus passage. It is assumed that a fraction δ of the total initial flow rate Q_0 passes through the population of large pores and the remaining fraction $(1 - \delta)$ passes through the population of small pores. The initial virus retention, LRV_0 , may be obtained as follows:

$$LRV_0 = -\log\{[S_s(1 - \delta)Q_0 + S_l\delta Q_0]/Q_0\} = -\log(S_l\delta) \quad (1)$$

where S_s is the sieving coefficient for the combined flow through the small pores ([virus filtrate from small pores]/[virus charge]) and S_l is the sieving coefficient for the combined flow through the large pores ([virus filtrate from large pores]/[virus filtrate]). Equation (1) may be simplified by assuming that the passage of virus through the small-pore population is zero (i.e. $S_s = 0$). If the small pores become selectively plugged, the total flow through the membrane (Q) decreases and instantaneous LRV may be obtained in terms of LRV_0 , Q and Q_0 . By the rearrangement of eqn (1) (under the assumption that fluid flow through the large pores remains unchanged):

$$LRV = -\log(S_l\delta Q_0/Q) = LRV_0 - \log(Q_0/Q) \quad (2)$$

Eqn (2) indicates that a 90% decline in flow from its initial value gives a 10-fold reduction in virus retention.

It should be noted that this model is based on the assumption that the larger-pore population remains unaffected. This is not unreasonable for selective plugging by virus-sized particles, but would not be expected in the case of adsorptive fouling, because these pores would also be constricted by protein build-up. In this case, these 'large' pores may behave as 'small' pores. Thus more sophisticated modelling is warranted to predict the behaviour of adsorption-fouled filters.

For virus filters that may have more than one physical membrane layer, an analogous equation may be derived in terms of n , the number of functionally independent 'effective layers' in the membrane. It is assumed that, like 'theoretical plates' in a column, a membrane that is thicker than one or a few virus diameters may be divided into layers, each of which is capable of functioning independently to sieve particles.

The number of 'effective layers' would be a function of the thickness of individual membranes and the additive effect of membrane layers. Thus:

$$\text{LRV}_0 = -\log(S_0\delta)^n \quad (3)$$

Eqn (3) assumes that the overall sieving coefficient for a multilayer membrane is equal to the product of the sieving coefficients of the individual layers. Assuming that each membrane is plugged to the same extent, the overall virus retention may be obtained as follows:

$$\text{LRV} = -\log(S_0\delta Q_0/Q)^n = \text{LRV}_0 - n \log(Q_0/Q) \quad (4)$$

For otherwise similar conditions, the reduction in LRV predicted by eqn (4) for a multilayer filter will be higher than for a single-layer filter. However, at the same time, LRV_0 would be higher for thicker or multilayer membranes, raising the starting point from which the LRV would fall. This is explained by the fact that the resistance of a multilayer filter is equal to the sum of the individual layer resistances, whereas the sieving is the product of individual layer sieving coefficients.

Again, it is worth noting that a membrane that has a highly interconnected pore structure may behave like many 'effective membranes' in series. Although an extreme interpretation of this model predicts that the LRV can decrease to zero with only a minor reduction in flow rate, in practice this would require an infinitely thick membrane and is not achievable. In the case of Viresolve NFP filters, there are three physical layers, and hence n is at least 3 [10].

Materials and methods

Proteins

Human IgG was purchased from SeraCare Life Sciences (Oceanside, CA, U.S.A.) as a 5% stock solution (catalogue no. HS-475). IgG was diluted to a final concentration of 1 mg/ml in Difco FA buffer, pH 7.2 (Fisher Scientific, Pittsburgh, PA, U.S.A.). Before its use in filtration experiments, the IgG was further purified by filtration through Millipore's Viresolve Prefilter (catalogue no. SSPVA4ONB9), stored at 4 °C and used within 48 h. In the experiments shown in Figure 5, IgG was further processed as follows: human IgG (SeraCare Life Sciences) at 5 mg/ml was filtered through a 25.4 cm (10 in) Opticap™ with Viresolve Prefilter media (Millipore catalogue no. KSPV01FF1), then through a single-layer Viresolve/180 membrane housed in a 293-mm-diameter stainless-steel holder (Millipore catalogue no. YY3029316). The single layer of Viresolve/180 membrane was intended to remove protein aggregates and provide a primarily monomeric stream. A sample of this filtrate was tested for throughput through a Viresolve NFP membrane as a control at the beginning and at the end of a series of experiments. The Viresolve NFP throughput did

not decline, indicating that aggregates were not forming over the course of the series of experiments. The filtrate was adjusted to a final concentration of 1 mg/ml in PBS (pH 7.2).

Manufacturing-grade BSA was purchased from Serologicals Corporation (Norcross, GA, U.S.A.) as a 30% stock solution (catalogue no. 81-070-5) and diluted to the appropriate concentration in Difco FA buffer (pH 7.2).

mAb (monoclonal antibody) and TPs were developmental proteins from Eli Lilly (Indianapolis, IN, U.S.A.). For experiments using mAb, the test buffer was 25 mM Tris and 500 mM ammonium sulphate (pH 7.8–8.2), and for experiments using TP the buffer was 10 mM citrate and 175 mM NaCl (pH 5.9–6.3). These proteins were used at the concentrations indicated in the Figure legends.

All buffer and protein solutions were processed through a 0.22 µm-pore-size GP Express® Plus filter (catalogue no. SCGPU02RE; Millipore Corporation, Bedford, MA, U.S.A.) before use in virus retention experiments.

Preparation of virus stocks

ΦX-174 in sodium tetraborate buffer was purchased from Promega Corporation (Madison, WI, U.S.A.) with a stock titre of approx. 6×10^{12} p.f.u. (where p.f.u. stands for plaque-forming units)/ml. MMV (murine minute virus) stocks were grown in the Mammalian Virology Laboratory at Millipore Corporation using source stocks purchased from the American Type Culture Collection (Manassas, VA, U.S.A., catalogue no. VR-1346). The 324K cells (simian-virus-40-transformed human newborn kidney cells, provided by Dr P. Tattersall, Department of Laboratory Medicine, Yale University Medical School, New Haven, CT, U.S.A.) were infected with MMV at a multiplicity of infection of 0.001 and cultured in high glucose DMEM (Dulbecco's minimum essential medium) supplemented with 1% FBS (fetal bovine serum), 2 mM L-glutamine and 100 µM non-essential amino acids (Invitrogen, Carlsbad, CA, U.S.A.). Infected cells were incubated at 37 °C in 5% CO₂ until approx. 80% of the cells showed CPE (cytopathic effect). High-titre MMV stocks were then prepared using a modification of the method described by Previsani et al. [14]. The infected 324K cells were pelleted by low-speed centrifugation and resuspended at approx. 1×10^7 cells/ml in TE buffer (10 mM Tris/HCl and 1 mM EDTA, pH 8.7). After three freeze-thaw cycles, the cell suspension was clarified by spinning at 18000 g, for 20 min at 4 °C and the virus pelleted by spinning the supernatant at 83 100 g, for 18 h at 4 °C. The virus pellet was then resuspended in a small volume of TE buffer (≈0.25 ml per T-150 flask of infected cells) and stored in aliquots at –80 °C. Stock titres were approx. 10^{10} TCID₅₀ (50% tissue-culture infectious dose)/ml.

Preparation of Mock virus stocks

The MMV mock preparation was made from 324K cells maintained in DMEM + 1% FBS. Cells were frozen and

thawed three times in DMEM + 1% FBS, the supernatant was clarified by low-speed centrifugation and then stored in aliquots at -80°C . This procedure mimics the preparation of a crude MMV stock. In some experiments, the protein feedstream was spiked with the MMV mock preparation at 1% (v/v).

Polystyrene beads

Polystyrene beads were purchased from Bangs Laboratories (Fishers, IN, U.S.A.) in two sizes: 21 nm (catalogue no. PS02N/6565) and 90 nm beads (catalogue no. PS02N/3947). The beads were aliquoted into pH 7.2 PBS with 500 p.p.m. Pluronic F-68 and gently mixed.

Virus assays

The infectious titre of $\Phi\text{X-174}$ in samples was determined using the plaque assay method. Samples were serially diluted (10-fold), mixed with *Escherichia coli* HF4714 in liquid overlay agar and the mixture was poured on to solid agar plates (three or four plates for each dilution). Plates were then incubated overnight at 37°C and plaques were counted the next day. Titres were calculated from plates containing between 30 and 300 plaques by multiplying the number of plaques and the dilution factor, then dividing by the sample volume plated.

The infectious titre of MMV in test samples was determined by TCID₅₀ assays performed in the Mammalian Virology Laboratory (Millipore Corporation). Samples were serially diluted (5-fold) using high glucose DMEM with 1% FBS. Monolayers of 324K cells in 24-well plates were infected using at least eight 0.25 ml replicates of the appropriate dilution of sample. Plates were then incubated at 37°C with 5% CO₂ and individual wells were observed for CPE on days 10–12 after infection. Virus titres in TCID₅₀/ml were calculated by the Kärber method using the equation:

$$\log \text{TCID}_{50}/\text{ml} = d + [f(s - 0.5)] + 0.6 \quad (5)$$

where d is the log₁₀ (starting dilution), f the log(dilution factor) and s the sum of the proportion of CPE positive wells at each dilution.

Virus-retention experiments

The filtration methodology included the collection and assay of fractions taken over the course of filtration. The test filters in all experiments were Optiscale™-25 devices with Viresolve NFP membrane (catalogue no. SVPVA25NB9; Millipore Corporation). Viresolve NFP contains three layers of a composite PVDF membrane. The membrane structure is shown in Figure 1.

Filtration experiments were run at 30 lbf/in² (1 lbf/in² = 6.9 kPa) and ambient temperature unless otherwise noted. Devices were first wetted with test buffer (without protein) at 30 lbf/in² for 5–10 min and the buffer flow rate during the

final minute of wetting was calculated. The buffer flow rate was considered to be the initial flow rate during test runs. The system was then depressurized, the buffer drained from the pressure vessel and the challenge solution (protein spiked with virus) was added. Challenge solutions were processed through a 0.22 μm -pore-size GP Express Plus filter before being added to the pressure vessel. The system was then pressurized to 30 lbf/in² and the volume of filtrate was measured every minute for the duration of the test using precision timers and balances (process fluid density was assumed to be 1 g/ml). The initial flow rate was used to calculate target flow rates at specific levels of flow decline (generally 25, 50, 75 and 90% decline from the initial flux). Samples were collected directly from the filter devices 2–3 min after start of each run (initial sample) and at the target flow reductions. Samples collected directly from the filter devices are referred to as 'grab samples'. These samples measure the virus retention capability of the filter at the time of sample collection, as opposed to pooled filtrate samples typically collected in validation studies that measure virus retention of the filter over the course of an entire run. Samples were also taken from the spiked challenge solution at the beginning of each run and the end of the run (hold control). The virus titre in the challenge solution was calculated as the average of the titre in the initial challenge sample and the hold control. LRVs were calculated as follows: $\log(\text{virus titre in the challenge solution}/\text{virus titre in the filtrate grab samples})$.

Electron microscopy

Samples were sputter-coated using an Au/Pd target before scanning-electron-microscope analysis. Images were acquired using the Topcon DS-130-C scanning electron microscope (Aspex Instruments LLC, Delmont, PA, U.S.A.).

Results

The primary aim of this study was to gain a better understanding of the process parameters that affect virus retention and ultimately to develop a scientific foundation for validation of virus filters. Previous studies with virus filters suggested that virus retention as well as liquid flow declines over the course of a filtration run [10, 11]. Since two liquid flow-related variables are changing over the course of filtration (the volume processed and the flow rate), initial work focused on understanding which of these parameters is more closely associated with virus retention.

Flow decline and bacteriophage breakthrough

Bacteriophages are commonly used as surrogates of mammalian viruses to test the efficiency of a membrane

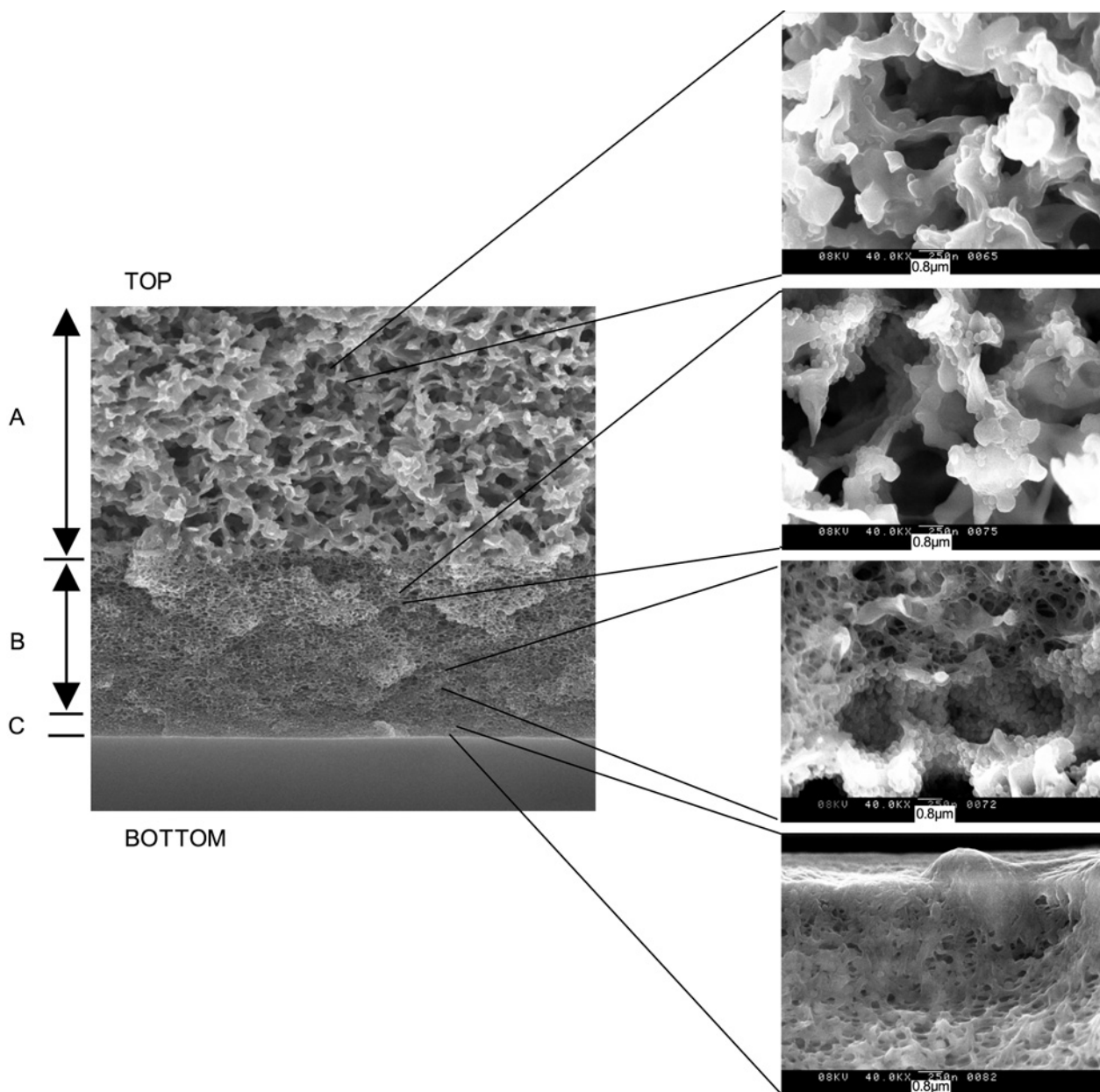


Figure 1 Scanning-electron-micrograph image of a single layer of Viresolve/180 membrane

The left image corresponds to a height of approx. 30 μm and shows the tight UF (ultrafilter) layer at the bottom, the transition zone above the UF and the substrate. The images on the right have higher magnifications of each area with scale bars as shown. Caking of 90 nm beads can be seen in the second and third images from the bottom on the right.

separation for viral reduction [8,9,15–18]. In order to demonstrate the suitability of the bacteriophage $\Phi\text{X-174}$ as a surrogate for mammalian parvoviruses, an experiment was performed in which either $\Phi\text{X-174}$ or MMV was spiked into a model feedstream (human IgG). It was then processed through Viresolve NFP filters, and the LRVs were measured at various points during the run (Figure 2). Both viruses are robustly cleared by Viresolve NFP filtration at

the start of the run (e.g. flow decay < 25–50%). The general trends of LRV decline as a function of volume processed (Figure 2A) or percentage flow decline (Figure 2B) were very similar with the two viruses. Thus $\Phi\text{X-174}$ appears to approximate MMV's behaviour in virus-filtration experiments.

Clearance of viruses at the start of a filtration operation has been previously established to be generally insensitive

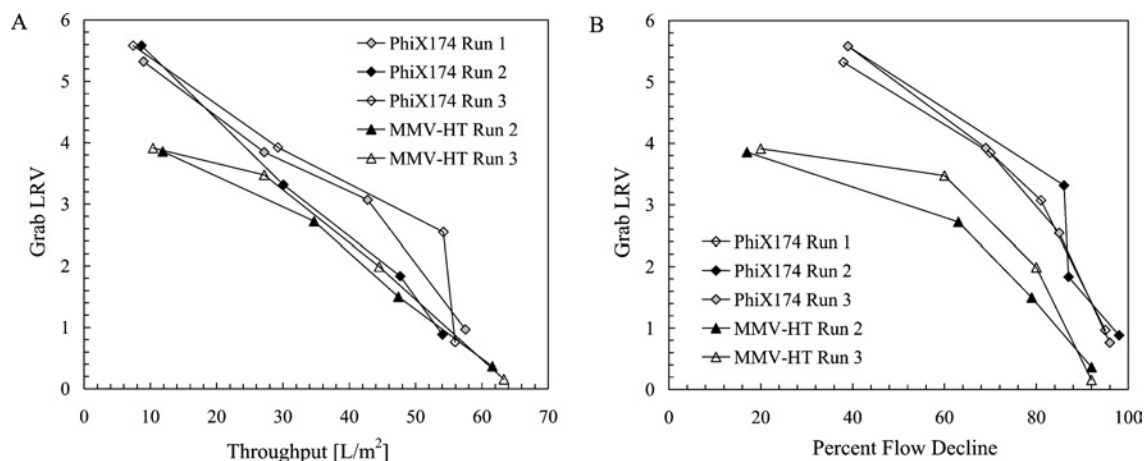


Figure 2 Comparison of ΦX-174 and MMV clearance with NFP filters

Human IgG (1 mg/ml) was spiked with ΦX-174 at 1×10^6 p.f.u./ml or MMV at 1×10^6 TCID₅₀/ml. One grab sample was taken at min 2 of each run. Additional grab samples were collected between 60 and 95 % flow decline. Data are plotted as grab LRV versus throughput (A) or grab LRV versus percentage flow decline (B). Pool samples were also assayed at 79–87 % flow decline in all experiments; for ΦX-174, the pool LRVs were 3.8, 3.1 and 3.4; for MMV the values were 3.1 and 3.2.

to the effects of common operating variables such as trans-membrane pressure, filter batch, initial flow rate and protein concentration [8,9]. To identify variables that can impact flow decline over extended processing, experiments where either the protein concentration was varied (Figures 3A–3C) or different batches of IgG diluted from the same stock were performed (Figure 3D). In each case, the challenge solution was spiked with bacteriophage ΦX-174 at a concentration of 10^6 – 10^7 p.f.u./ml. Figures 3(A)–3(D) show the LRV data (for grab samples) plotted against throughput (litre/m²) for a wide range of conditions including different proteins, protein concentrations and buffer conditions. In all cases, the LRV decreased as a function of throughput, but the slopes of the LRV versus throughput curves were found to be system-specific and varied as a function of all parameters tested in the present study, including protein, protein concentration and IgG preparation history.

As all of these variables can impact overall flow and throughput over time, the LRV data obtained in this study with Viresolve NFP was also plotted against flow decline (Figure 4). The data from different runs collapse on to a single curve, demonstrating that LRV decay correlates well with flow decline. This indicates that flow decline is more predictive of LRV than volumetric throughput. The data also fit the type of curve generated by the pore-plugging model described by eqn (4). The curve has a shape that is exponential, although the optimal power factor (i.e. n) of the curve varies between 3 and 5 in Figures 4(A)–4(D). It is likely that this reflects additional variables between the process feeds (e.g. impact of adsorption, aggregate populations) that adds complexity to the flow decay behaviour not captured by eqn (4).

Fouling mechanisms

Fouling mechanisms appear to add complexity to the flow-decay-versus-LRV relationship; thus it would be useful to define the mechanism operative in validation experiments vis-à-vis process scale filtration and understand the impact of different fouling mechanisms on virus filter performance. Potential mechanisms for filter fouling include direct deposition of a cake of very large particulates on the surface of the membrane, adsorption of proteins or other molecules throughout the pore structure and plugging of pores by particles and protein aggregates [19]. In reality, all three happen to some extent during filtration. Fouling by cake formation is thought to occur with particles that are too large to even pass through the entrance of the pores, whereas with pore plugging, particles enter the pores but become lodged inside. Additional experiments to assess the mechanism of fouling during viral-reduction studies were performed to determine if these mechanisms of filter fouling differentially impact LRV during extended filtration.

In the first series of experiments, shown in Figure 5, the membrane was fouled with model (polystyrene) beads of various sizes chosen to cause membrane fouling by either pore plugging (21 nm), cake formation (90 nm) or a monomeric IgG solution to cause pore constriction by adsorption. It is hypothesized that the monomeric IgG is adsorptive because it does cause flow decay even though all aggregates were removed by filtration before the experiment. Monomeric IgG is generally regarded to adsorb to surfaces and has a hydrodynamic diameter (≈ 10 nm) that is smaller than the pores in a Viresolve NFP filter. The concentration of beads was 2×10^{11} and 7×10^{11} ml⁻¹ for the 21 and 90 nm

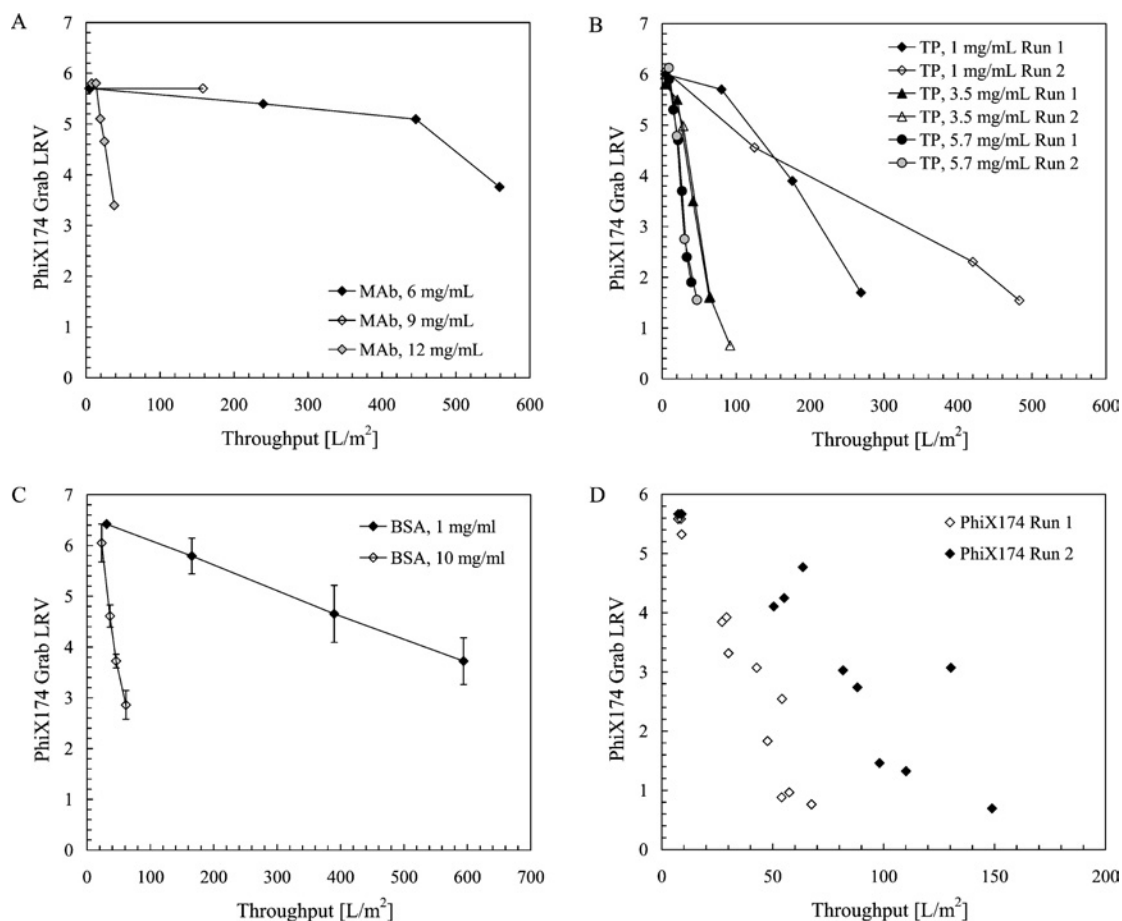


Figure 3 Relationship between LRV and throughput for several different filtrations performed under a range of process conditions

(A, B) mAb and TP were diluted to the concentrations indicated. Each protein solution was spiked with Φ X-174 at approx. $3\text{--}6 \times 10^6$ p.f.u./ml. (C) Manufacturing grade BSA was diluted to the indicated concentration in Difco FA buffer (pH 7.2) and spiked with Φ X-174 at approx. 1×10^7 p.f.u./ml. Data points represent the average for four experiments. (D) Human IgG (1 mg/ml) was spiked with Φ X-174 at 1×10^6 p.f.u./ml. The results are from experiments run on two separate days with two different batches of human IgG made from the same stock. Data points shown are from three devices for each experiment.

beads respectively. The model prediction (eqn 4, assuming $LRV_0 = 6.5$ and $n = 3$) is also plotted.

The data in Figure 5 suggest that the impact of fouling on the LRV flow-decline profile depends on the mechanism of fouling. It is seen that the model with the three-layer assumption adequately describes the behaviour of the 21 nm latex beads. As predicted, build-up of a cake of 90 nm beads above the virus-retentive layer causes gradual flow restriction but has no impact on phage breakthrough. This is consistent with their having no impact on pore-size distribution in the virus-sieving portion of the membrane that they cannot enter. Eqn (4) underpredicts the LRV decrease with the adsorptive monomeric IgG solution at 90% flow decline. IgG adsorption causes a similar exponential LRV-versus-flow-decay curve indicative of an even more profound pore-size distribution shift. In this case, eqn (4) fits if $n = 5$ or 6. More sophisticated modelling is warranted. It should be noted, however, that flow decay is again far more predictive

of LRV after adsorptive fouling compared with total throughput. Thus it is a superior filtration endpoint for extended processing regardless of the mechanism of flow decay (e.g. plugging, adsorption and caking).

Bead penetration in the filter membranes was investigated by electron microscopy. Visual observation of scanning images of the membrane cross section taken after fouling of the membrane (Figure 1) revealed that the 90 nm beads were predominantly trapped at the interface between the UF transition zone and the substrate, confirming the cake formation mechanism. The 21 nm beads, Φ X-174, and the monomer IgG were too small to be visualized by electron microscopy.

Virus stock impurities

In the next series of experiments, we carried out tests to simulate the conditions observed in validation studies. Virus validation studies are often performed with mammalian

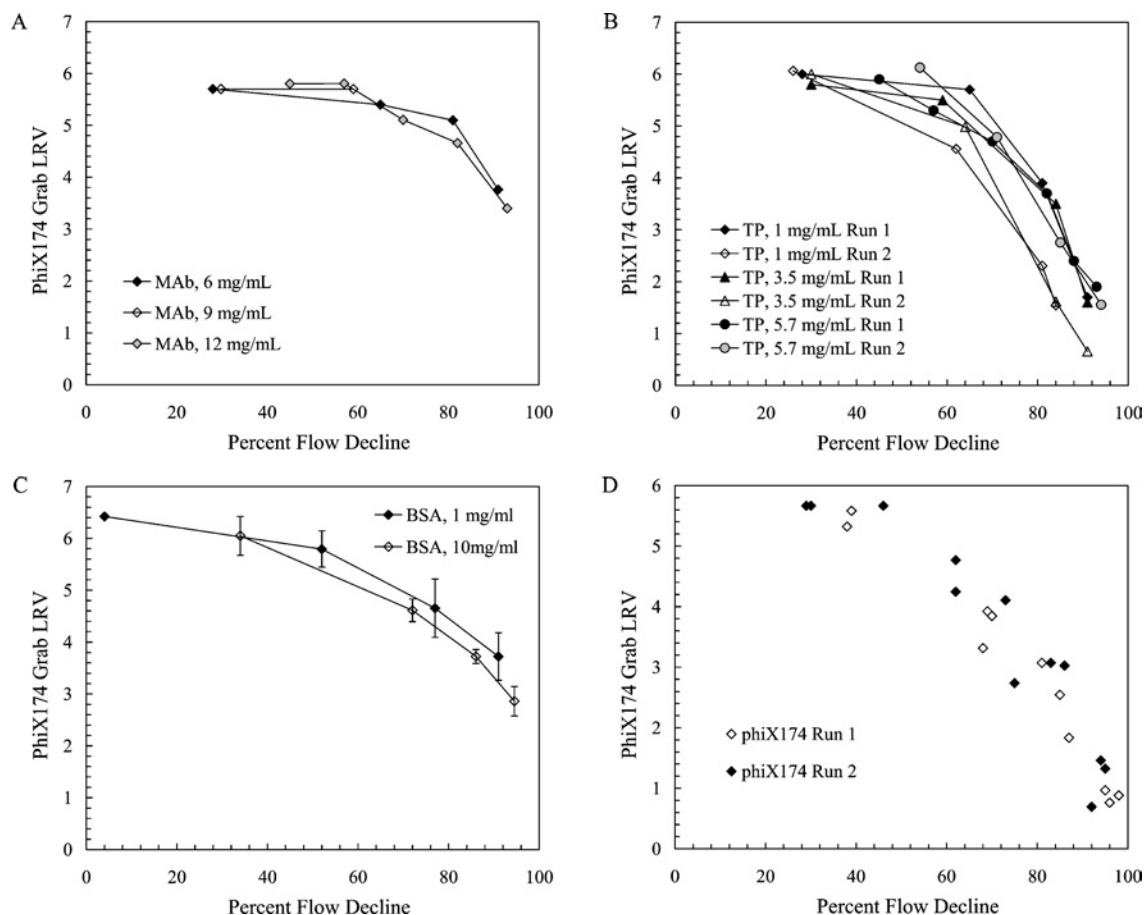


Figure 4 Relationship between LRV and flow decline for several different filtrations performed under a range of process conditions

The Φ X-174 LRV data from Figure 3 were plotted against percentage flow decline for TPs, mAbs, BSA and IgG solutions. The LRV declines with percentage flow decline in a feed stream-independent manner, as the different runs collapse on to the TP, mAB, BSA and IgG solutions.

virus stocks purchased from vendors that may or may not attempt to concentrate the virus or remove potential caking, adsorptive or plugging impurities such as media components, host cell proteins or DNA. In addition, some of these preparations may contain aggregates of host cell proteins or the viruses themselves that would also plug pores. All of these adsorptive and particulate contaminants probably contribute to the flow decline in a typical virus-filter validation study. In fact, when a crude virus preparation is spiked into a highly purified protein solution, these components in all likelihood contribute far more to the filter fouling than the actual process solution itself.

To determine if such impurities cause flow decline consistent with our pore-plugging model, typical validation feedstocks were simulated by spiking IgG feed with cell culture supernatant from 324K cells, the host for MMV (MMV mock preparation). As a comparison, the IgG feed was spiked with gradient-purified Φ X-174, which has only trace levels of non-phage contaminants that can potentially con-

tribute to filter fouling. The data are plotted in Figure 6 as LRV versus flow decline. In accordance with eqn (4), the results for the two fouling agents (proteins, cell lysate constituents and cell culture supernatant) are indistinguishable, indicating that flow decline caused by impurities that would be present in a typical validation feedstock is predictive of viral clearance under process conditions.

Discussion

Plugging of pores by aggregates and other particles in the virus-spike model solutions and the resultant decrease in LRV has important implications for both manufacturers of biological products and suppliers of filters. Validation is performed in laboratory-scale virus-reduction studies using process intermediates and model virus preparations. Standard practice for these studies is to use representative solutions and conditions at a flux that is scaled down from

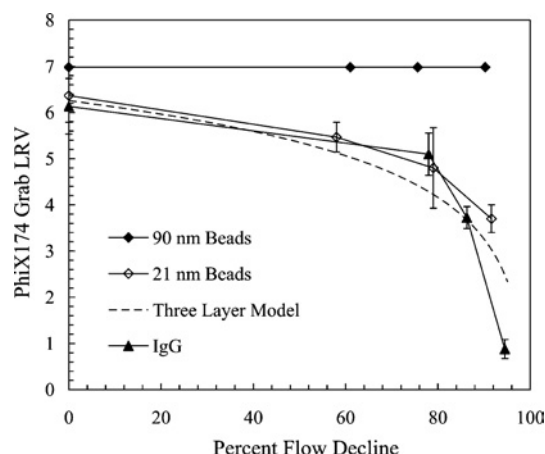


Figure 5 LRV versus flow decline for membranes fouled with pore plugging 21 nm beads, cake-forming 90 nm beads or 1 mg/ml adsorptive fouling monomer IgG

The model prediction adequately describes the behaviour of the 21 nm latex beads but overpredicts the LRV with IgG. Build-up of a cake of 90 nm beads above the filter membrane causes gradual flow restriction but has no impact on phage breakthrough.

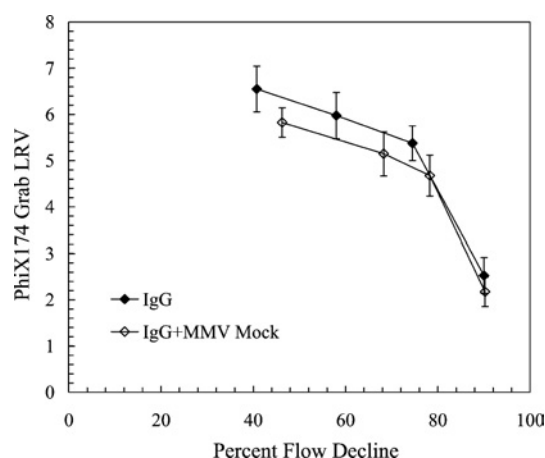


Figure 6 PhiX174 LRV versus percentage flow decline for IgG and IgG spiked with cell culture supernatant from 324K cells, the host for MMV

IgG was diluted to a final concentration of 1 mg/ml in Difco FA buffer (pH 7.2), and then purified by filtration through a diatomaceous-earth filter (prototype filter; similar to Millipore part no. KA1HC10FF1). The IgG was then spiked with either Φ X-174 alone at 10^7 p.f.u./ml or Φ X-174 plus MMV mock preparation at 1% (v/v). Each data point is the average for four experiments. The results for the two fouling agents (protein and cell culture supernatant) are indistinguishable, indicating that flow decline caused by impurities that will be present in a typical validation feedstock is predictive of viral clearance. The surrogate virus spike does not lead to higher LRVs than those observed when only IgG is filtered.

the large-scale process in the large-scale process. In the current practice, the total fluid throughput, corrected for the total filter membrane area (litre/m²), is matched with the small-scale studies and is defined as the filtration endpoint. Considering the data presented in Figure 3, it is evident that

this approach may be flawed, since, in each case, the plot is highly systemspecific. For the mAb, TP and BSA, the LRV at any given throughput varies significantly depending on the concentration of the protein (Figures 3A–3C). In the case of human IgG, experiments were performed on two different days using two batches of protein (both at 1 mg/ml) made from the same concentrated stock solution (Figure 3D). In the experiments shown in Figure 3(D) there were significant differences in throughput on the two test days, which resulted in very different LRV versus throughput curves. For example, at 60 litre/m² in run 1 the grab LRVs were approx. 1.0 whereas in run 2, grab samples taken at the same throughput level had LRVs of 4.0–5.0.

The graphs in Figure 4, however, indicate that the data are brought together on to a single curve when plotted against the percentage flow decline, as would be predicted by eqn (4). Thus, in contrast with the data plotted as LRV versus throughput in Figure 3(D), Figure 4(D) shows a very good correlation between LRV and flow decline in the two IgG experiments. The explanation for the lower LRVs at 60 litre/m² in experiment 1 is that the flow decline was much higher at this point than in experiment 2. The same holds true for the other proteins in which the concentration was varied (compare Figures 3A–3C with Figures 4A–4C). In each case, LRV is more closely related to flow decline than throughput. Taken together, these data indicate that flow decline is a more appropriate endpoint for virus filtration experiments compared with throughput.

Importantly, contamination with spiked surrogate virus material does not lead to higher LRVs than those observed when only protein, IgG, is filtered (Figure 6). These results support the use of flow decline as the basis for defining filtration endpoint, whether the flow decline is caused by fouling due to the protein product itself or due to the virus spike.

The slope of the LRV-versus-flow-decline curves, although relatively independent of initial process conditions like protein content, differ somewhat between model proteins. It is possible that plugging versus adsorptive effects may vary between biopharmaceutical proteins. This argues that the flow decay versus LRV relationship should be characterized at small scale when processes for new biopharmaceutical products are undergoing development or validation. However, once established for a new product, our results argue that this relationship will not be impacted by factors such as total throughput or protein concentration if these are changed during process development.

Past experimental observations have reported the phenomenon of virus breakthrough as a function of litres of filtrate/m² of filter area for a number of filters: Asahi Planova I-BMM and Pall DV20 [11,12]. The present paper demonstrates for the first time that flux decline is a unifying parameter characterizing breakthrough.

Conclusions

This paper provides experimental data and a new model to gauge the useful endpoint of membrane filtration used for viral clearance. We found that under a variety of filtration conditions, plots of LRV versus flow decline were strikingly similar and agreed well with predictions of a model of LRV as a function of membrane fouling. This is in contrast with graphs of the same data where LRV was plotted against throughput in which LRV varied significantly depending on the conditions. Currently, validation by throughput/unit area is the common approach for validation studies. However, our results suggest that, at least for some filters, performing virus-filter validation studies with a flow decline endpoint may be more appropriate than using throughput as the endpoint. Universal acceptance of this argument would be strengthened by similar studies in other filters. Some variation, 2–3, in the extent of LRV decline was noted between different model proteins, but validation studies are always biopharmaceutical product-specific and such differences would be measured in these or development studies anyway. Our experiments also demonstrated that only when a flow decline endpoint far beyond that acceptable in commercial manufacturing (i.e. > 50–75%) is achieved is the LRV value below that expected in a process scale operation. Finally, spiking studies done in the presence of fouling material from a mammalian virus stock (as simulated with a mock virus preparation) gave similar small virus retention to studies in which the protein feed was the only plugging agent (simulating actual process conditions). These data support the idea that the fouling mechanisms operative in a validation study are representative of process conditions.

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